Achieving Envy-Freeness with Limited Subsidies under Negative Dichotomous Valuations

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Abstract

We study the problem of allocating indivisible *chores* among agents in a fair manner, with subsidies. Envy-free allocations of indivisible items are not guaranteed to exist, but envy-freeness can be achieved by additionally providing some subsidy to the agents; in the chores setting this subsidy is naturally read as compensation paid to the agents who absorb the workload. We study the subsidy required under *negative dichotomous* valuations, i.e., among agents whose marginal disutility for any chore is either zero or one: $v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\}$ for every agent i, every $S \subseteq M$ and every $g \in M \setminus S$. This is the exact mirror of the dichotomous goods model of Barman et al. [BKNS22], who prove that there a per-agent subsidy of 0 or 1 always suffices. We prove that the same guarantee holds for chores.

Existing work already gives a bounded subsidy: negative dichotomous valuations are doubly monotone under the two-sided normalisation, so the reduction of Kawase et al. [KMS+24] yields $n - 1$ per agent and $n(n-1)/2$ in total. The question is whether the far stronger $\{0, 1\}$ guarantee of [BKNS22] survives the passage from goods to chores, a factor of n below that baseline. The Halpern–Shah characterisation of envy-freeable allocations [HS19] transfers verbatim, since it assumes nothing about the sign or the structure of the valuations, and it delivers an *integral* minimum subsidy vector here; the matching lower bound of $n - 1$ transfers as well, so the bound in question is tight.

We prove that bound, for every number of agents: every negative dichotomous instance admits an envy-free solution with a subsidy of 0 or 1 per agent, hence at most $n - 1$ in total, computable in polynomial time in the value-oracle model, and under no assumption on the cost functions beyond binary marginals — not additivity, not submodularity, no relation between different agents' cost functions, and no bound on the number of chores. The proof takes an envy-free *partial* allocation, produced by the algorithm of Tao et al. [TWYZ23] and leaving at most $n - 1$ chores unassigned, and completes it: the residual chores go to distinct agents inside the strongly connected component at which that algorithm halts, and the paid set is the backward closure of those recipients under the relation "agent i values her own bundle exactly as she values agent j's". Envy-freeness of the completion is verified directly, so the argument does not appeal to the characterisation of envy-freeable allocations.

1 Introduction

Discrete fair division studies the allocation of resources that cannot be fractionally assigned among agents with individual preferences, and sits at the interface of mathematical economics and computer science [BCE+16, End17]. The classical fairness criterion is *envy-freeness* [Fol67, Var74]: no agent should prefer the bundle assigned to another agent over her own. For indivisible items an envy-free allocation need not exist, as the standard one-item two-agent instance shows, and a large body of work therefore studies relaxations such as envy-freeness up to one good (EF1) [Bud11, LMMS04].

A second and complementary line of work retains *exact* envy-freeness and buys off the residual envy with money. Each agent i receives, in addition to her bundle, a subsidy $p_i \geq 0$ supplied by a third party, and one asks for an allocation together with a subsidy vector under which every agent weakly prefers her own bundle-plus-subsidy to that of every other agent. The question is how much money this requires. Maskin [Mas87] answered it for unit-demand agents with $m = n$: under the scaling in which no agent values any single item above 1, a total subsidy of $n - 1$ suffices. Halpern and Shah [HS19] moved to the multi-demand setting with arbitrary m, gave the characterisation of *envy-freeable* allocations that the whole line now runs on, showed that a subsidy of $(n-1)m$ is necessary and sufficient in the worst case when the allocation is handed to us, and conjectured that $n - 1$ suffices when we may choose the allocation. Brustle et al. [BDN+20] proved that conjecture in the stronger per-agent form — one dollar to each agent suffices for additive valuations, with the allocation simultaneously EF1 and balanced — and showed that for general monotone valuations $2(n-1)$ per agent suffices, so that the total subsidy is $O(n^2)$ and, remarkably, *independent of the number of items*. Barman et al. [BKNS22] then improved the monotone bound by a factor of n on the dichotomous subclass: when every marginal $v_i(S \cup \{g\}) - v_i(S)$ lies in $\{0, 1\}$, there is an allocation whose accompanying subsidy is exactly 0 or 1 for each agent, hence at most $n - 1$ in total, computable in polynomial time in the value-oracle model. Their bound assumes neither additivity nor submodularity nor even subadditivity, and it is tight: with n agents and a single unit-valued good, $n - 1$ agents must each be paid 1.

Chores. Every result above is about *goods*. This paper asks the mirrored question, in which every item is a *chore*: an object that an agent would rather not hold, so that acquiring it weakly decreases her utility. Chores are not an exotic variant. Shift rotations, on-call duty, teaching and refereeing load, committee service, maintenance tasks and the liabilities attached to an estate are all allocation problems in which the items are burdens rather than prizes, and in all of them money is already the standard instrument of repair — as overtime pay, hardship allowance, or a course-release budget. If anything the subsidy model is more natural for chores than for goods, since compensating somebody for taking on unwanted work needs less justification than paying somebody for the privilege of not receiving a prize.

Concretely, we study *negative dichotomous* valuations: for every agent i, every $S \subseteq M$ and every $g \in M \setminus S$,

$$v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\} .$$

Equivalently, each marginal $v_i(S \cup \{g\}) - v_i(S)$ is 0 or -1: every item is a chore for every agent, and taking on one more chore costs an agent either nothing or exactly one unit. This is the exact mirror of the model of [BKNS22], and we impose no further structure — in particular the valuations need not be additive, submodular, or subadditive.

Binary marginals are the natural model whenever a task either lands on an agent as a fresh unit of burden or is absorbed by a commitment she has already made. An engineer already staffing an on-call window absorbs a second incident in that window at no extra cost;

a referee already reading for a venue absorbs one more paper from the same batch; a driver already routed through a district absorbs an extra stop along the way. Whether a given chore is free depends in an arbitrary way on the rest of the agent's assignment, which is exactly why the model must tolerate non-additive and even non-subadditive valuations, and it is the counterpart of the scheduling example used by [BKNS22] to motivate dichotomous valuations for goods.

The baseline and the lower bound. Two facts delimit the problem, and we state them explicitly so that the contribution can be measured against them.

On the one hand, existence of a bounded subsidy is not in question. Negative dichotomous valuations are doubly monotone — every item is a chore for every agent, which is a special case of every item being a good or a chore for each agent — and they satisfy the normalisation $|v_i(S \cup \{e\}) - v_i(S)| \leq 1$ that [KMS+24, §2] imposes. The reduction of Kawase et al., which converts any EF1 allocation into an envy-free one, therefore applies off the shelf and yields a subsidy of at most $n-1$ per agent and $n(n-1)/2$ in total [KMS+24, Theorem 1]. Two details make the containment work and are easy to get wrong. Their normalisation is two-sided, so it bounds chore disutility as well as value; and the EF1 they require is the two-sided form of Definition 5, which removes one item from *either* bundle, not the goods-only form of [HS19] — the two differ precisely when chores are present. An EF1 allocation is guaranteed to exist and to be computable in polynomial time for doubly monotone valuations, so the reduction has an input; for that step [KMS+24, §2] cites [LMMS04] and [BSV21], and the citation matters, because the envy-cycle argument of [LMMS04] does *not* port to chores unaltered and it is [BSV21] that supplies the repair.

On the other hand, the lower bound of the goods model survives intact.

Example 1 (The $n-1$ lower bound transfers)**.** Take n agents and $m = n-1$ chores, with $c_i(S) = |S|$ for every agent i — identical, binary, additive costs, which are in particular negative dichotomous. Every complete allocation leaves at least one agent empty-handed. Under the balanced allocation, in which $n-1$ agents hold one chore each and one agent holds nothing, each of the $n-1$ loaded agents envies the empty agent by exactly 1 and no other envy is present, so the minimum subsidy assigns 1 to each loaded agent and 0 to the empty one, for a total of $n-1$. No allocation does better: consider any complete allocation $\mathcal{A} = (A_1, \dots, A_n)$, and write $s_i = |A_i|$. Since there are only $n-1$ chores, $\sum_{i=1}^{n} s_i = n-1$, and therefore at least one agent, say h, receives no chore: $s_h = 0$. Suppose that a subsidy vector $p \in \mathbb{R}_+^n$ makes the allocation envy-free. In cost form, envy-freeness requires

$$c_i(A_i) - p_i \leq c_i(A_j) - p_j \qquad \text{for all } i, j.$$

Comparing every agent i with the empty-bundle agent h, we obtain

$$s_i - p_i = c_i(A_i) - p_i \leq c_i(A_h) - p_h = -p_h.$$

Hence

$$p_i \geq s_i + p_h \geq s_i \qquad \text{for every } i.$$

Summing over all agents gives

$$\sum_{i=1}^{n} p_i \geq \sum_{i=1}^{n} s_i = n-1.$$

So the target statement is precisely the analogue of [BKNS22, Theorem 4].

Theorem 2 (Main result)**.** *For every instance with negative dichotomous valuations there is an envy-free solution* $(\mathcal{A}, p)$ *with* $p \in \{0,1\}^n$*, hence with total subsidy at most* $n-1$*; and it can be computed in polynomial time given value-oracle access.*

Example 1 shows this bound is tight. Between the $n-1$ per agent that [KMS⁺24] already gives and the $\{0,1\}$ per agent asked for lies a factor of n in the total, exactly the gap that [BKNS22] closed on the goods side.

Our contribution. We establish the statement above, for every number of agents (Theorem 21). The proof takes as input an envy-free *partial* allocation — the algorithm of Tao et al. [TWYZ23] produces one in polynomial time, leaving at most $n-1$ chores unassigned — and completes it: the residual chores are assigned to distinct agents inside the strongly connected component at which that algorithm halts, and the paid set is the backward closure of those recipients under the equality relation $c_i(X_i) = c_i(X_j)$. Envy-freeness of the completion is verified directly, so no appeal to the characterisation of envy-freeable allocations is required. Section 2 fixes the model and records what transfers from the goods theory unchanged; Section 3 contains the proof.

Related Work. The subsidy line for goods is described above. Within it, the dichotomous subclass has attracted separate attention: Halpern and Shah [HS19] settle binary additive valuations with a total subsidy of $n-1$, Goko et al. [GIK⁺21] obtain the analogous bound for binary submodular valuations together with a strategy-proof mechanism, and Barman et al. [BKNS22] subsume both by dropping every structural assumption beyond binary marginals. Dichotomous preferences are themselves a well-studied restriction in social choice [BMS05, BEF21]. On the computational side, Caragiannis and Ioannidis [CI21] show that minimising the subsidy is NP-hard and give a fully polynomial-time approximation scheme for a constant number of agents, and Narayan et al. [NSV21] study the welfare cost of achieving envy-freeness with transfers.

The paper closest to the present setting is that of Kawase et al. [KMS⁺24], the only work in the envy-freeness-with-subsidy line that admits chores at all. Working with doubly monotone valuations under the two-sided normalisation, it decouples the two jobs that [BDN⁺20] performed together: rather than constructing a specific allocation and showing that it is cheap to subsidise, it takes *any* EF1 allocation as input and bounds the subsidy needed to convert it, obtaining $n-1$ per agent and $n(n-1)/2$ in total. Since EF1 allocations are known to exist for doubly monotone valuations, the wider class comes along at no cost. That reduction is what makes the present question well-posed rather than open-ended, and it is the baseline we are trying to beat.

The complementary repair — no money, weaker fairness notion — is pursued for binary valuations by Bu et al. [BSY23], who show that EFX allocations always exist and can be computed in polynomial time for dichotomous goods valuations, thus extending the matroid-rank result of Babaioff et al. [BEF21]. Under binary valuations one therefore has a clean dichotomy on the goods side: either pay 0 or 1 per agent and obtain exact envy-freeness [BKNS22], or pay nothing and obtain EFX [BSY23]. This paper establishes the first branch for chores.

The chores side. Work on indivisible chores has largely developed alongside rather than inside the subsidy line, and the two have met less often than one might expect. Where money has been brought to chores, it has been in the service of *proportionality* rather than envy-freeness: the question studied is the total subsidy needed to bring every agent down to her proportional share of the burden, for which the best bounds under additive disutilities normalised to at most 1 per chore are $n/4$ and then $n/3 - 1/6$, recently obtained together with Pareto optimality by Garg et al. [GSW25]. That is a different fairness notion and a different accounting; it does not bound the subsidy needed for envy-freeness.

The no-money branch, by contrast, has a direct chore analogue of [BSY23]. Tao et al. [TWYZ23] study chores with binary marginals and show that for binary *additive* costs an allocation that is simultaneously EFX and Pareto optimal always exists and is computable in polynomial time. So on the chores side the second half of the binary dichotomy — pay nothing, obtain EFX — is established at least for additive costs, while the first half, pay 0 or 1 per agent and obtain exact envy-freeness, is what this paper establishes. That money is genuinely needed here is not obvious a priori but is known: Bhaskar et al. [BSV21] show that deciding whether an exactly envy-free allocation of chores exists is NP-complete already for binary additive costs, and give a polynomial-time EF1 algorithm for chores and for doubly monotone instances.

Where the two lines have met is on the *additive* side, and recently. Lu et al. [LMS26] prove the goods-and-chores analogue of [BDN+20]: for additive utilities in which each item is a good for some agents and a chore for others, with every $|v_i(g)| \le 1$, a subsidy of one dollar per agent suffices, the bound is tight, and the allocation is computable in polynomial time — hence $n - 1$ in total. This settles the additive chore problem completely.

It does not settle ours, and the reason is exactly the reason [BKNS22] was not a corollary of [BDN+20]. Dichotomous valuations are not additive and additive valuations are not dichotomous; the two classes are incomparable. The four corners are

	goods	chores
additive	[BDN+20]	[LMS26]
dichotomous	[BKNS22]	**this paper**

each of the four giving one dollar per agent and $n - 1$ in total. Note also that [LMS26] does not displace [KMS+24] for the class studied here: for valuations that are dichotomous but not additive, the $n - 1$ per agent of [KMS+24] was the best previously known bound, and it is that bound which the present result improves by a factor of n.

We are not aware of any prior treatment of envy-freeness with subsidy for chores under dichotomous costs.

2 Notation and Preliminaries

We study the allocation of m indivisible items among n agents, with subsidies. Write $N = [n]$ for the set of agents and M for the set of items, $|M| = m$. Each agent $i \in N$ has a valuation $v_i : 2^M \to \mathbb{R}$ with $v_i(\emptyset) = 0$, and we represent an instance by the tuple $\langle N, M, \{v_i\}_{i \in N} \rangle$. Algorithms operate in the standard *value-oracle* model: for each v_i we assume only an oracle that returns $v_i(S)$ for a queried set $S \subseteq M$, and running time is measured in n, m and the number of oracle calls.

An *allocation* $\mathcal{A} = (A_1, \ldots, A_n)$ is an ordered tuple of pairwise disjoint *bundles*, with A_i assigned to agent i. The allocation is *complete* if $\bigcup_{i \in N} A_i = M$ and *partial* otherwise. The ordering matters throughout: much of the theory below concerns permuting a fixed family of bundles among the agents, and for a permutation $\sigma \in \mathbb{S}_n$ we write $\mathcal{A}_\sigma := (A_{\sigma(1)}, \ldots, A_{\sigma(n)})$ for the allocation in which agent i receives $A_{\sigma(i)}$. The utilitarian welfare of $\mathcal{A}$ is $\mathrm{SW}(\mathcal{A}) = \sum_{i \in N} v_i(A_i)$.

Definition 3 (Envy-Freeness)**.** *An allocation $\mathcal{A} = (A_1, \ldots, A_n)$ is* envy-free *(EF) iff $v_i(A_i) \geq v_i(A_j)$ for all agents $i, j \in N$.*

Envy-free allocations of indivisible items need not exist, so we allow a third party to supplement the allocation with money. Agents have quasi-linear utilities: agent i receiving bundle A_i and subsidy p_i enjoys utility $v_i(A_i) + p_i$, with money entering linearly and at the same exchange rate for every agent.

Definition 4 (Envy-Free Solution)**.** *An allocation $\mathcal{A} = (A_1, \ldots, A_n)$ and a subsidy vector $p = (p_1, \ldots, p_n) \in \mathbb{R}^n_+$ constitute an* envy-free solution *$(\mathcal{A}, p)$ iff $v_i(A_i) + p_i \geq v_i(A_j) + p_j$ for all $i, j \in N$.*

An allocation $\mathcal{A}$ is *envy-freeable* if some $p \in \mathbb{R}^n_+$ makes $(\mathcal{A}, p)$ an envy-free solution; this is a property of the allocation alone. Following [BKNS22] we write $\mathrm{M}(p) := \{i \in N \mid p_i \geq p_j \text{ for all } j \in N\}$ for the set of maximally subsidised agents.

We will occasionally need the standard relaxation of envy-freeness. Note that two inequivalent forms of it appear in this literature: the goods-only form of [HS19], which removes one item from the envied bundle, and the two-sided form used by [KMS+24], which removes one item from *either* bundle. They coincide when all items are goods and differ once chores are present, and only the two-sided form supports the doubly monotone results. We adopt the two-sided form.

Definition 5 (EF1, two-sided form)**.** *An allocation $\mathcal{A}$ is* envy-free up to one item *(EF1) iff for all $i, j \in N$ there exists $X \subseteq A_i \cup A_j$ with $|X| \leq 1$ such that $v_i(A_i \setminus X) \geq v_i(A_j \setminus X)$.*

2.1 Negative Dichotomous Valuations

Definition 6 (Negative Dichotomous Valuation)**.** *A valuation $v_i : 2^M \to \mathbb{R}$ with $v_i(\emptyset) = 0$ is* negative dichotomous *iff*

$$v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\} \qquad \text{for every } S \subseteq M \text{ and every } g \in M \setminus S,$$

equivalently iff every marginal $\Delta^+_i(S, g) := v_i(S \cup \{g\}) - v_i(S)$ lies in $\{-1, 0\}$.

Every item is thus a chore for every agent: v_i is non-increasing, so $v_i(S) \geq v_i(T)$ whenever $S \subseteq T$. We stress that nothing further is assumed — the valuations need not be additive, submodular, or subadditive — which is the generality at which [BKNS22] works on the goods side.

It is often more convenient to work with the non-negative *cost form*

$$c_i := -v_i, \qquad \text{so} \qquad c_i(\emptyset) = 0, \quad c_i(S) \leq c_i(T) \text{ for } S \subseteq T, \quad c_i(S \cup \{g\}) - c_i(S) \in \{0, 1\}.$$

Definition 7 (Dichotomous cost function)**.** *A set function* $c : 2^M \to \mathbb{R}$ *is* dichotomous *if* $c(\emptyset) = 0$ *and every marginal* $c(S \cup \{g\}) - c(S)$ *lies in* $\{0, 1\}$.

Monotonicity is not a separate hypothesis: marginals in $\{0, 1\}$ are non-negative, so a dichotomous c is automatically non-decreasing. We nonetheless list it above because it is the property most often used directly.

The correspondence $v_i \leftrightarrow c_i$ is a bijection between negative dichotomous valuations and dichotomous cost functions in the sense of [BKNS22]: the class we study is exactly the pointwise negation of the class they study. In cost form an envy-free solution is the requirement $c_i(A_i) - p_i \le c_i(A_j) - p_j$ for all i, j, read as "my own burden net of my compensation is no worse than what I perceive of yours". We use whichever of v_i and c_i is clearer locally and always state which.

2.2 The Envy Graph and the Halpern–Shah Characterisation

For an allocation $\mathcal{A}$, the *envy graph* $G_{\mathcal{A}}$ is the complete weighted directed graph on vertex set N in which the arc (i, j) carries weight

$$w_{\mathcal{A}}(i, j) \;:=\; v_i(A_j) - v_i(A_i) \;=\; c_i(A_i) - c_i(A_j),$$

the envy that agent i has towards agent j. The weight of a directed path P is $w_{\mathcal{A}}(P) = \sum_{(i,j)\in P} w_{\mathcal{A}}(i, j)$, and $\ell_{\mathcal{A}}(i)$ denotes the maximum weight of any path in $G_{\mathcal{A}}$ starting at i, the empty path of weight 0 included.

The two results below are the engine of the entire subsidy line. Both are stated by Halpern and Shah [HS19] for additive valuations, but their proofs use only the arc weights and permutations of the bundles — neither additivity nor monotonicity nor any sign condition on v_i enters — and [BKNS22] invokes them in exactly this generality for (non-additive) dichotomous valuations. They therefore apply verbatim to negative dichotomous valuations, and we use them without further comment.

Theorem 8 ([HS19])**.** *For any allocation* $\mathcal{A} = (A_1, \dots, A_n)$*, the following are equivalent.*

(i) *$\mathcal{A}$ is envy-freeable.*

(ii) *$\mathcal{A}$ maximises utilitarian welfare across all reassignments of its own bundles among the agents: for every permutation σ over N,* $\sum_{i\in N} v_i(A_i) \ge \sum_{i\in N} v_i(A_{\sigma(i)})$.

(iii) *The envy graph $G_{\mathcal{A}}$ has no positive-weight directed cycle.*

Theorem 9 ([HS19])**.** *For any envy-freeable allocation $\mathcal{A}$, the subsidy vector p^* given by $p^*_i := \ell_{\mathcal{A}}(i)$ realises an envy-free solution $(\mathcal{A}, p^*)$, and $p^*_i \le p_i$ for every $i \in N$ and every p such that $(\mathcal{A}, p)$ is an envy-free solution. It can be computed in strongly polynomial time by running an all-pairs longest-path computation on $G_{\mathcal{A}}$.*

Condition (ii) of Theorem 8 has the standing consequence that an envy-freeable allocation can always be manufactured from an arbitrary one: fix the bundles and reassign them according to a maximum-weight perfect matching between agents and bundles. What is at issue is never the existence of an envy-freeable allocation, only the size of the subsidy that Theorem 9 then charges for it.

Specialising to our model gives the following, which we use throughout.

Observation 10 (Integrality)**.** *Let $\{v_i\}_{i\in N}$ be negative dichotomous. Then $v_i(S) \in \mathbb{Z}_{\le 0}$ for every i and every $S \subseteq M$; every arc weight $w_{\mathcal{A}}(i,j)$ is an integer; and for every envy-freeable allocation $\mathcal{A}$ the minimum subsidy vector satisfies $p^* \in \mathbb{Z}^n_{\ge 0}$.*

Proof Fix i and $S = \{g_1, \ldots, g_k\}$. Telescoping from $v_i(\emptyset) = 0$ along any ordering of S writes $v_i(S)$ as a sum of k marginals, each of which lies in $\{-1, 0\}$ by Definition 6; hence $v_i(S) \in \mathbb{Z}$ and $-k \le v_i(S) \le 0$. Consequently each arc weight $w_{\mathcal{A}}(i,j) = v_i(A_j) - v_i(A_i)$ is a difference of integers, and the weight of any directed path, being a finite sum of arc weights, is an integer. By Theorem 9, $p_i^* = \ell_{\mathcal{A}}(i)$ is the maximum such weight over paths starting at i, and it is non-negative because the empty path has weight 0. □

Observation 10 is what makes the target statement — a subsidy vector in $\{0,1\}^n$ — a statement about a *bound* rather than about rounding: once p^* is known to be integral, showing $p_i^* \le 1$ for every i is the same as showing $p^* \in \{0,1\}^n$. The same observation underlies the corresponding step in [BKNS22], where the arc weights of a dichotomous instance are likewise integers.

A second normalisation costs nothing and removes a clause from the target.

Proposition 11 (Some agent is always unpaid)**.** *For every envy-freeable allocation $\mathcal{A}$ there is an agent h with $p_h^* = 0$. Consequently, if $p^* \in \{0,1\}^n$ then $\sum_{i\in N} p_i^* \le n-1$.*

Proof By Theorem 8(iii) the envy graph $G_{\mathcal{A}}$ has no positive-weight cycle, so a walk of maximum weight may be taken simple and $\ell_{\mathcal{A}}$ is finite; it is non-negative because the empty path qualifies. Choose a path P attaining $\max_i \ell_{\mathcal{A}}(i)$ and let h be its final vertex. If $\ell_{\mathcal{A}}(h) > 0$ there is a positive-weight walk Q leaving h, and P followed by Q is a walk from P's start of weight $w_{\mathcal{A}}(P) + w_{\mathcal{A}}(Q) > w_{\mathcal{A}}(P)$, contradicting the maximality of P among walks. Hence $\ell_{\mathcal{A}}(h) = 0$. The second claim is immediate: at most $n-1$ coordinates of p^* can equal 1. □

Proposition 11 justifies the word "hence" in the target statement of Section 1: the bound on the *total* is not an independent requirement but a consequence of the per-agent one. For the pointwise-minimal subsidy the target is therefore the single-clause statement

$$\exists \mathcal{A} \; : \quad \max_{i\in N} \ell_{\mathcal{A}}(i) \;\le\; 1.$$

3 Unit Subsidies for Negative Dichotomous Valuations

This section proves the main theorem: for every number of agents n and every negative dichotomous instance there is an allocation admitting an envy-free solution with a subsidy of 0 or 1 per agent, at most $n-1$ in total, computable in polynomial time. Beyond Definition 7 nothing is assumed of the cost functions — not additivity, not submodularity, not any relation between different agents' cost functions — and the number of chores is unrestricted.

The argument takes as input an envy-free *partial* allocation produced by the algorithm of Tao, Wu, Yu and Zhou [TWYZ23], and completes it. That algorithm, its Theorem 5.1, and the three update rules recorded in Section 3.1 are due to [TWYZ23] and are restated here for reference; the completion of Section 3.2 onwards, and every lemma and theorem after Lemma 14, are ours.

We work in cost form throughout, so an envy-free solution (Definition 4) is the requirement

$$c_i(A_i) - p_i \;\le\; c_i(A_j) - p_j \qquad \text{for all } i, j \in N, \tag{1}$$

and a partial allocation is envy-free (Definition 3) when

$$c_i(A_i) \;\le\; c_i(A_j) \qquad \text{for all } i, j \in N. \tag{2}$$

For $S \subseteq M$ and $e \in M \setminus S$ write $c_i(e \mid S) := c_i(S \cup \{e\}) - c_i(S) \in \{0, 1\}$ for a marginal, and for a partial allocation $\mathcal{A}$ write $R(\mathcal{A}) := M \setminus \bigcup_{i \in N} A_i$ for its *residue*.

Observation 12. *For every $i \in N$ and all $S, T \subseteq M$,*

$$c_i(S) < c_i(T) \quad \Longrightarrow \quad c_i(S) \le c_i(T) - 1.$$

Moreover $c_i(S) \le c_i(T)$ whenever $S \subseteq T$.

Proof By Observation 10 every c_i is integer valued, so a strict inequality between two of its values is a gap of at least 1. For monotonicity, enumerate $T \setminus S = \{f_1, \ldots, f_q\}$ and telescope: $c_i(T) - c_i(S) = \sum_{u=1}^{q} c_i\bigl(f_u \mid S \cup \{f_1, \ldots, f_{u-1}\}\bigr)$, a sum of terms in $\{0, 1\}$ by Definition 7, hence non-negative. □

3.1 The terminal state of the partial-allocation algorithm

Theorem 13 ([TWYZ23], Theorem 5.1)**.** *For every n and all dichotomous cost functions $c_1, \ldots, c_n$ there is a polynomial-time algorithm returning a partial allocation X that is envy-free in the sense of* (2) *and satisfies $|R(X)| \le n - 1$.*

The proof below uses not only the statement of Theorem 13 but the state in which the algorithm of [TWYZ23] halts, so we record that algorithm. It maintains a partial allocation $X = (X_1, \ldots, X_n)$, envy-free throughout, together with the *equality graph*

$$G = (N, E), \qquad (i, j) \in E \iff i \ne j \;\text{ and }\; c_i(X_i) = c_i(X_j), \tag{3}$$

and repeats the following three rules in the stated order, halting only when none applies.

(R1) If some $e \in R(X)$ and some $i \in N$ satisfy $c_i(e \mid X_i) = 0$, assign e to i.

(R2) If some $e \in R(X)$ and some arc $(i, j) \in E$ lying on a directed cycle of G satisfy $c_i(e \mid X_j) = 0$, rotate the bundles along that cycle and assign e to i.

(R3) Otherwise select a strongly connected component S of G with no arc of E leaving S. If $|R(X)| \ge |S|$, assign one residual chore to each agent of S; otherwise halt.

The loop runs while $R(X) \ne \emptyset$, so the only exit with $R(X) \ne \emptyset$ is the final clause of (R3).

For the remainder of this section fix the following data: let X be the partial allocation at which the algorithm halts, let $R := R(X)$ and $r := |R|$, and, when $r \ge 1$, let S be the component selected by (R3) at the halting step, with G and E as in (3) evaluated at that state.

Lemma 14. *Suppose $r \ge 1$. Then*

(a) $r < |S|$;

(b) $c_i(e \mid X_i) = 1$ *for every* $i \in N$ *and every* $e \in R$;

(c) $c_i(e \mid X_j) = 1$ *for every arc* $(i, j) \in E$ *with* $i, j \in S$ *and every* $e \in R$;

(d) $c_i(X_i) < c_i(X_j)$ *for every* $i \in S$ *and every* $j \in N \setminus S$.

Proof Since $r \geq 1$, the algorithm halted at the final clause of (R3); hence neither (R1) nor (R2) applied at the halting state, and $|R| < |S|$, which is (a).

(b) If $c_i(e \mid X_i) = 0$ for some $i \in N$ and some $e \in R$, rule (R1) would apply. Marginals lie in $\{0, 1\}$ by Definition 7, so the value is 1.

(c) Let $(i, j) \in E$ with $i, j \in S$. Since S is strongly connected there is a directed path in G from j to i; taking one with no repeated vertex and appending the arc (i, j) yields a directed cycle of G through (i, j). If $c_i(e \mid X_j) = 0$ for some $e \in R$, rule (R2) would apply to that arc and that cycle. Hence the marginal is not 0, so it is 1.

(d) Let $i \in S$ and $j \in N \setminus S$. No arc of E leaves S, so $(i, j) \notin E$, i.e. $c_i(X_i) \neq c_i(X_j)$ by (3). Envy-freeness (2) gives $c_i(X_i) \leq c_i(X_j)$, so the inequality is strict. □

3.2 The completion

Assume $r \geq 1$ and write $R = \{e_1, \ldots, e_r\}$. By Lemma 14(a), $r < |S|$, so we may fix r distinct agents

$$T = \{t_1, \ldots, t_r\} \subseteq S,$$

chosen arbitrarily. Define $\mathcal{A} = (A_1, \ldots, A_n)$ by

$$A_i := \begin{cases} X_{t_k} \cup \{e_k\}, & i = t_k \ \ (1 \leq k \leq r), \\ X_i, & i \in N \setminus T. \end{cases} \tag{4}$$

The t_k are distinct and each chore of R is used exactly once, so the bundles of $\mathcal{A}$ are pairwise disjoint and cover M: $\mathcal{A}$ is a complete allocation. Members of T are called *recipients*.

Definition 15 (Subsidy set)**.** *Let* $P \subseteq N$ *be the smallest set satisfying*

(P1) $T \subseteq P$; *and*

(P2) if $i \in N \setminus S$ *and* $(i, j) \in E$ *for some* $j \in P$*, then* $i \in P$.

Equivalently, P *is obtained by initialising* $P := T$ *and repeatedly adjoining any* $i \in N \setminus S$ *having an arc of* E *into the current set, until no further agent can be adjoined; the process terminates because* N *is finite and* P *only grows. Define*

$$p_i := \begin{cases} 1, & i \in P, \\ 0, & i \in N \setminus P. \end{cases} \tag{5}$$

Lemma 16. $P \setminus T \subseteq N \setminus S$*: every agent of* P *that is not a recipient lies outside* S.

Proof Let $P_0 := T$ and let P_{u+1} be P_u together with every $i \in N \setminus S$ having an arc of E into P_u, so that $P = \bigcup_{u \geq 0} P_u$. We show $P_u \setminus T \subseteq N \setminus S$ by induction on u. For $u = 0$ the set $P_0 \setminus T$ is empty. For the step, every agent adjoined in passing from P_u to P_{u+1} lies in $N \setminus S$ by construction. □

Lemma 17. *$p \in \{0,1\}^n$ and $\sum_{i \in N} p_i = |P| \leq n-1$.*

Proof That $p \in \{0,1\}^n$ with $\sum_i p_i = |P|$ is immediate from (5). By Lemma 16, $P \subseteq T \cup (N \setminus S)$, and this union is disjoint because $T \subseteq S$; hence $|P| \leq |T| + |N \setminus S| = r + n - |S|$. By Lemma 14(a), $r \leq |S| - 1$, so $|P| \leq n - 1$. □

Lemma 18. *Let $i \in S$, let $t \in T$ and let $e \in R$. Then*

$$c_i\big(X_t \cup \{e\}\big) \;\geq\; c_i(X_i) + 1.$$

Proof Note $t \in T \subseteq S$. Envy-freeness (2) gives $c_i(X_i) \leq c_i(X_t)$.

If $c_i(X_i) < c_i(X_t)$, then $c_i(X_t) \geq c_i(X_i) + 1$ by Observation 12, and $c_i(X_t \cup \{e\}) \geq c_i(X_t)$ by monotonicity, again Observation 12.

Otherwise $c_i(X_i) = c_i(X_t)$. If $i = t$ then $c_i(X_t \cup \{e\}) = c_i(X_i) + c_i(e \mid X_i) = c_i(X_i) + 1$ by Lemma 14(b). If $i \neq t$ then $(i,t) \in E$ by (3), with $i, t \in S$, so $c_i(e \mid X_t) = 1$ by Lemma 14(c) and therefore $c_i(X_t \cup \{e\}) = c_i(X_t) + 1 = c_i(X_i) + 1$. □

3.3 Envy-freeness of the completion

Theorem 19. *Let X, R, S be as in Section 3.1 with $r \geq 1$, let $\mathcal{A}$ be as in (4) and let p be as in (5). Then $(\mathcal{A}, p)$ is an envy-free solution.*

Proof Fix $i, j \in N$; we verify (1), that is $c_i(A_i) - p_i \leq c_i(A_j) - p_j$. Throughout, $X_j \subseteq A_j$ for every j by (4), so by monotonicity (Observation 12) and (2),

$$c_i(A_j) \;\geq\; c_i(X_j) \;\geq\; c_i(X_i). \tag{6}$$

Note also that $T \subseteq P$, so $i \notin P$ implies $i \notin T$ and hence $A_i = X_i$; likewise $j \notin P$ implies $A_j = X_j$ and $p_j = 0$. Since N is the disjoint union of T, $P \setminus T$ and $N \setminus P$, the following three cases are exhaustive.

Case 1: $i \in T$**.** Then $p_i = 1$ and, by Lemma 14(b), $c_i(A_i) = c_i(X_i) + c_i(e \mid X_i) = c_i(X_i) + 1$, so $c_i(A_i) - p_i = c_i(X_i)$. Also $i \in T \subseteq S$.

(1a) $j \in T$. Then $p_j = 1$ and $A_j = X_j \cup \{e\}$ for some $e \in R$, so $c_i(A_j) \geq c_i(X_i) + 1$ by Lemma 18. Hence $c_i(A_j) - p_j \geq c_i(X_i)$.

(1b) $j \in P \setminus T$. Then $p_j = 1$ and $A_j = X_j$. By Lemma 16, $j \notin S$; since $i \in S$, Lemma 14(d) gives $c_i(X_i) < c_i(X_j)$, hence $c_i(X_j) \geq c_i(X_i) + 1$ by Observation 12. Hence $c_i(A_j) - p_j = c_i(X_j) - 1 \geq c_i(X_i)$.

(1c) $j \in N \setminus P$. Then $p_j = 0$ and $A_j = X_j$, so $c_i(A_j) - p_j = c_i(X_j) \geq c_i(X_i)$ by (2).

Case 2: $i \in P \setminus T$**.** Then $p_i = 1$ and $A_i = X_i$, so $c_i(A_i) - p_i = c_i(X_i) - 1$. For every j we have $p_j \leq 1$ and $c_i(A_j) \geq c_i(X_i)$ by (6), hence $c_i(A_j) - p_j \geq c_i(X_i) - 1 = c_i(A_i) - p_i$.

Case 3: $i \in N \setminus P$**.** Then $p_i = 0$ and $A_i = X_i$, so $c_i(A_i) - p_i = c_i(X_i)$.

(3a) $j \in N \setminus P$. Then $p_j = 0$ and $A_j = X_j$, so $c_i(A_j) - p_j = c_i(X_j) \geq c_i(X_i)$ by (2).

(3b) $j \in T$. Then $p_j = 1$ and $A_j = X_j \cup \{e\}$ for some $e \in R$; we claim $c_i(A_j) \geq c_i(X_i) + 1$. If $i \in S$ this is Lemma 18. If $i \notin S$, suppose $c_i(X_i) = c_i(X_j)$. Then $i \neq j$, since $j \in T \subseteq P$ while $i \notin P$; so $(i, j) \in E$ by (3), and $j \in P$, so (P2) of Definition 15 places $i \in P$ — contradicting $i \notin P$. Hence $c_i(X_i) < c_i(X_j)$, so $c_i(X_j) \geq c_i(X_i) + 1$ by Observation 12, and $c_i(A_j) \geq c_i(X_j)$ by monotonicity. In either case $c_i(A_j) - p_j \geq c_i(X_i)$.

(3c) $j \in P \setminus T$. Then $p_j = 1$ and $A_j = X_j$; we claim $c_i(X_j) \geq c_i(X_i) + 1$. If $i \notin S$, suppose $c_i(X_i) = c_i(X_j)$. Then $i \neq j$, since $j \in P$ while $i \notin P$; so $(i, j) \in E$, and (P2) places $i \in P$, a contradiction. Hence $c_i(X_i) < c_i(X_j)$. If instead $i \in S$, then $j \notin S$ by Lemma 16, and Lemma 14(d) gives $c_i(X_i) < c_i(X_j)$. In either case Observation 12 yields $c_i(X_j) \geq c_i(X_i) + 1$, so $c_i(A_j) - p_j = c_i(X_j) - 1 \geq c_i(X_i)$.

In every sub-case $c_i(A_j) - p_j \geq c_i(X_i) = c_i(A_i) - p_i$ in Cases 1 and 3, and the inequality was shown directly in Case 2. □

3.4 The EF1 property of the completion

Corollary 20. *The complete allocation $\mathcal{A}$ constructed in Section 3.2 is EF1. More precisely, every non-recipient $i \in N \setminus T$ is envy-free in $\mathcal{A}$, while for every recipient $t_k \in T$, removing the residual chore e_k assigned to t_k eliminates all of t_k's envy.*

Proof Recall that the partial allocation $X = (X_1, \ldots, X_n)$ is envy-free, and that $X_j \subseteq A_j$ for every $j \in N$. First, let $i \in N \setminus T$. Then $A_i = X_i$. Hence, for every $j \in N$, envy-freeness of X and monotonicity of c_i give

$$c_i(A_i) = c_i(X_i) \leq c_i(X_j) \leq c_i(A_j).$$

Thus i does not envy any agent in $\mathcal{A}$. Now let $i = t_k \in T$. By construction,

$$A_i = X_i \cup \{e_k\},$$

and hence

$$A_i \setminus \{e_k\} = X_i.$$

Therefore, for every $j \in N$,

$$c_i(A_i \setminus \{e_k\}) = c_i(X_i) \leq c_i(X_j) \leq c_i(A_j),$$

where the first inequality follows from envy-freeness of X and the second from monotonicity.

Thus, for every recipient, removing the single newly assigned chore from her own bundle eliminates her envy towards every other agent. Consequently, $\mathcal{A}$ is EF1. □

3.5 The main theorem

Theorem 21. *For every n and every instance $\langle N, M, \{v_i\}_{i \in N} \rangle$ with negative dichotomous valuations there exist a complete allocation $\mathcal{A}$ and a subsidy vector $p \in \{0, 1\}^n$ with $\sum_{i \in N} p_i \leq n - 1$ such that $(\mathcal{A}, p)$ is an envy-free solution. Moreover, the allocation $\mathcal{A}$ is EF1. Both $\mathcal{A}$ and p are computable in polynomial time given value-oracle access.*

Proof Pass to cost form, $c_i = -v_i$, which is dichotomous in the sense of Definition 7. Run the algorithm of Theorem 13 and let X, R, r and S be as fixed in Section 3.1.

If $r = 0$ then X is a complete allocation satisfying (2), so $(X, \mathbf{0})$ is an envy-free solution with $\sum_i p_i = 0 \leq n - 1$; being envy-free it is EF1, taking $X = \emptyset$ in Definition 5.

If $r \geq 1$, construct T, $\mathcal{A}$ and p as in Section 3.2. Then $\mathcal{A}$ is a complete allocation, Theorem 19 shows $(\mathcal{A}, p)$ is an envy-free solution, and Lemma 17 gives $p \in \{0,1\}^n$ with $\sum_i p_i \leq n - 1$. The allocation $\mathcal{A}$ is EF1 by Corollary 20.

For the running time, the algorithm of Theorem 13 is polynomial and returns X, R and S. Constructing $\mathcal{A}$ from (4) takes $r \leq n - 1$ assignments. The graph G of (3) has n vertices and is determined by the n^2 values $c_i(X_j)$, one oracle call each. Computing P is a backward reachability search in G from T restricted to vertices outside S, which takes $O(n^2)$ time. Hence the work after Theorem 13 is polynomial, and so is the whole procedure. □

By Example 1 the bound of Theorem 21 is tight: there are instances on which every complete allocation requires total subsidy at least $n - 1$.

Remark 22. Theorem 21 uses Theorem 13 together with the four properties of its halting state recorded in Lemma 14, which are read off the rules (R1)–(R3) of [TWYZ23] restated in Section 3.1; the statement of Theorem 13 alone does not supply them. Envy-freeness is verified directly from (1) in Theorem 19, so the argument does not invoke Theorem 8 or Theorem 9, and the subsidy exhibited need not be the pointwise-minimal one. Part (a) of Lemma 14 is used only in Lemma 17: the conclusion of Theorem 19 holds whenever $r \leq |S|$.

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